

XRTB: A Cross Reality Teleconference Bridge to incorporate 3D interactivity to 2D Teleconferencing

PRASANTH SASIKUMAR

psas598@aucklanduni.ac.nz
University of Auckland
NZ

HUIDONG BAI

University of Auckland
NZ

huidong.bai@auckland.ac.nz

MAX COLLINS

University of California Irvine
USA
mcollin2@uci.edu

MARK BILLINGHURST

University of Auckland
NZ



Figure 1: Asymmetrical virtual environments within existing teleconferencing system. Left most images show the view from Virtual environment as participants.

ABSTRACT

We present XRTeleBridge (XRTB), an application that integrates a Mixed Reality (MR) interface into existing teleconferencing solutions like Zoom. Unlike conventional webcam, XRTB provides a window into the virtual world to demonstrate and visualize content. Participants can join via webcam or via head mounted display (HMD) in a Virtual Reality (VR) environment. It enables users to embody 3D avatars with natural gestures and eye gaze. A camera in the virtual environment operates as a video feed to the teleconferencing software. An interface resembling a tablet mirrors the teleconferencing window inside the virtual environment, thus enabling the participant in the VR environment to see the webcam participants in real-time. This allows the presenter to view and

interact with other participants seamlessly. To demonstrate the system's functionalities, we created a virtual chemistry lab environment and presented an example lesson using the virtual space and virtual objects and effects.

CCS CONCEPTS

• **Human-centered computing** → *User interface management systems*; **Open source software**; *Systems and tools for interaction design*.

KEYWORDS

Remote Collaboration, Teleconference, Natural hand gestures, Virtual Reality

ACM Reference Format:

PRASANTH SASIKUMAR, MAX COLLINS, HUIDONG BAI, and MARK BILLINGHURST. 2021. XRTB: A Cross Reality Teleconference Bridge to incorporate 3D interactivity to 2D Teleconferencing. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3411763.3451546>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI '21 Extended Abstracts, May 8–13, 2021, Yokohama, Japan

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8095-9/21/05.

<https://doi.org/10.1145/3411763.3451546>

1 INTRODUCTION

In the current COVID-19 era, many people adapt to work, study, or play from home and communicate with others by using various video conferencing tools. However, some may discover the limitations of these tools when they try to present 3D spatial concepts or convey natural non-verbal communication cues through 2D camera feeds. For example, conducting a remote design review of a physical product with team members could be challenging via teleconference, which might not be a problem in a face-to-face meeting. A potential solution is to enable users to present themselves and their ideas in existing teleconferencing solutions in a more interactively and deliver much more information than 2D camera video.

VR is a great tool to provide rich interactive experience and VR based conferencing experiences have been developed to overcome the limitations discussed earlier. Applications such as Spatial.io(<https://spatial.io/>), (<https://altvr.com/>), Facebook Horizon(<https://www.oculus.com/facebook-horizon/>), Mozilla Hubs (<https://hubs.mozilla.com/>) etc. These work as a standalone platform, and participants have to use the corresponding application to collaborate remotely and lack integration to existing telecommunication platforms. This is the main motivation behind the development of XRTB.

XRTB solves this problem by restoring spatial cues like the depth information lost in video conferencing tools and enables remote collaboration to be more like face-to-face meetings. As shown in figure 1, we are able to accommodate virtual environment from within Zoom. We leverage VR and AR devices' ability to showcase natural non-verbal communication cues to provide a richer experience from within the traditional teleconference platform.

2 RELATED WORK

We reviewed previous implementations of remote MR collaboration and communication tools to understand design decisions. Implementations of systems where the participants were geographically separated and co-located were considered.

Some MR experiences are built to support collaboration between participants while they are co-located in the same physical space. Billinghurst and Kato express the importance of seamlessness in these face-to-face MR conditions and even found that participants of a collaborative face-to-face AR experience exhibit behaviors similar to interacting with physical objects when interacting with AR (digital) objects collaboratively [4]. This shows that there is an opportunity to learn from traditional face-to-face interactions when building virtual collaboration experiences. MR-mediated experiences where users must be co-located can offer benefits to users, such as using tools and objects that are not physically available to them and annotating their environments in ways that are not physically possible [7]. Construct3D gives users a collaborative environment to help understand mathematical concepts [8]. Using Construct3D, students could experiment with and explore with geometric models by exploiting their true three-dimensional nature. Kaufmann et al. found that Construct3D helped with spatial skills and will continue working with students to better build for student needs. An application of this nature could most effectively benefit the target population. These examples show that giving people the ability to visualize any data or object that can be sufficiently

modeled with colleagues is a great way to inspire new workflows, types of collaboration and demonstration, and ways of representing information and interacting with data. Keeping up with these examples, we design the system to have the ability to host co-located experiences. This way, more than one presenter can be part of the virtual environment to demonstrate complex tasks together in front of the camera.

2.1 Remote MR Collaboration systems

In scenarios where participants are not located in the same physical space, MR mediated experiences can help interact with one another. This could take the form of two participants joining each other in an entirely virtual environment from separate physical spaces by entering the same virtual reality. There are examples where remote participants interact with one another while they are each using HMD devices. There are also tools such as Mozilla Hubs(<https://hubs.mozilla.com/>) that allow participants to join a collaborative session via HMD or a traditional display device such as smartphones, tablets, or computers. These interactions can be synchronous (same time) or asynchronous (not at the same time) experiences, meaning participants can interact with each other in real-time or not.

Sodhi et al. developed a proof-of-concept system for synchronous AR collaboration[12]. Their system gave users the ability to see a remote partner's hand gestures and manipulations of digital objects in the physical world. This has implications for training, teaching, and demonstration of all sorts. Similarly, Gurevich et al. developed TeleAdvisor which gave remote collaborators the ability to annotate and point out things in the physical world to advise and train from afar [6]. Pointing out aspects of an environment to a remote user can be crucial in some circumstances and proves useful for users whose work lends itself to this [2]. Inspired by this, we incorporated spatial annotation as well as a drawing board.

Bai et al. looked into methods to improve participants' sense of co-presence while collaborating remotely using MR [1]. They found that they could significantly improve the sense of co-presence by allowing users to share gaze and gestures via digital overlays. Piumsomboon et al. also implemented a system to connect AR users to VR users, allowing them to collaborate by sharing hand gestures, gaze, and position [9]. Additionally, Previous studies have shown that representing body information improved social presence in remote collaboration, even if the representation is crude[11, 13]. This motivated the character representation detailed in section 3.3.

Chen et al. used the *Hololens* to help remote pairs collaborate in real-time, but unlike some other solutions, this involved remote participants joining from Skype sessions and "hitching" onto the view of the *Hololens* user [5]. This is a good example of asymmetrical remote environments still proving useful to remote participants. Remote participants who were hitching onto the HMD user's view could add annotations to the physical environment; it is important to keep in mind that asymmetry in task distribution can lead to differences in perspective when interacting in a virtual environment [10]. Building on top of these experiences, we incorporated our system to have avatar representation with hand gestures, annotating in space, and independent view point capabilities.

3 SYSTEM IMPLEMENTATION

The system is implemented as a standalone application that can work seamlessly alongside existing conferencing platforms. This ensures that our system does not require any change in user behavior for the participants. The presenter wears an MR device connected to a computer and joins the teleconferencing platform. We have tested our system with Zoom, Skype, Hangouts, Meet, and Facebook Messenger as teleconference clients. The system allows the presenter to switch the otherwise webcam feed to the VR environment's camera view, as shown in Figure 1. Meanwhile, the presenter can also keep the teleconferencing interface visible inside the virtual scene to keep up with the participants. Participants also have an option to join the virtual environment as an avatar and collaborate with the presenter. The prototype was built with Unity 3D Game Engine (2020.2.1f1) and tested on Oculus Quest (<https://www.oculus.com/quest/>) VR headset tethered to a Windows 10 PC with a link cable. The implementation can be broken down into three individual independent components: 1) Virtual Camera, 2) Desktop Mirroring, and 3) Virtual Environment. These components are discussed in detail in the following sections.

3.1 Virtual Camera

We developed a driver that registers itself as a virtual camera device. The driver creates an entry in the windows registry and identifies itself as a physical camera in the windows subsystem. In the unity project, a counterpart feeds the driver with texture information as a render texture. To achieve this, we used Unity's Scriptable Render Pipeline (Universal RP) to blit to the screen backbuffer using Graphics.Blit. In essence, Unity's virtual camera captures the scene and shares the view as render texture to a driver. This would be available as a video feed of a virtual camera and accessible to the teleconferencing applications.

Interactivity. : The presenter can control the position and rotation of the virtual camera in three different accessibility modes.

- **Manual control:** In manual control, the camera can be positioned by grabbing and moving the camera model.
- **Automatic control:** In automatic mode, the camera moves automatically and positions itself in front of the object the presenter is holding at a distance of 1 meter, which is configurable. If the presenter is not having any object, the camera positions itself 1.5m away, keeping the presenter in focus.
- **Hybrid Control:** The presenter can switch between the automatic and manual mode in the hybrid condition while presenting by pressing a button in the controller.

Preview. The virtual camera is represented by a 3D model of a classic camera as shown in Figure 2. The presenter can choose from a list of models, including DSLR, camcorder, mobile phone, and an action camera to represent the virtual camera. The models are provided with a grip handle for easy maneuver. And has a preview along both sides for the presenter. This is to make sure that the presenter has a preview regardless of the direction of the camera.

3.2 Desktop Mirroring

To enable the presenter to view the call participants, we provided a large tablet display inside the virtual environment, as shown

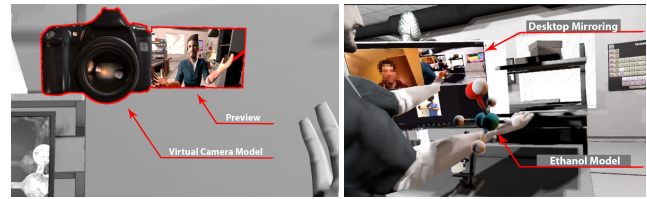


Figure 2: Left: Virtual Camera model and preview. View from a chemistry lab environment. Right: View from virtual camera where the presenter is explaining the 3D model while placing the conference window.

in Figure 2. To achieve this, we created a program that runs in parallel that captures the desktop and feeds to unity as a render texture, which is applied via material to the tablet surface. We also provided snap to fit positions throughout the virtual environment so that the presenter can easily place the tablet in comfortable viewing locations. For example, the presenter can leave the tablet near the whiteboard, and it will snap-fit to the board's side. The tablet interface can be interacted in three modes:

- **Manual control:** The presenter can position the tablet by grabbing and moving the tablet model.
- **Attach to Camera:** In the mode, the tablet is always attached to the virtual camera. Moving the virtual camera will move the tablet as well.
- **UI mode:** In this mode, the conference call view will always be available in the corner of the display as a UI element.

3.3 Virtual Environment and Character Representation

To test our concept, we created a relatively simple virtual environment considering scenarios that would help an academic presenter in a 3D space. Hence we incorporated a whiteboard where the presenter can draw using six different colored markers to discuss topics like equations and flowcharts. The virtual environment consists of three spaces, a lobby, an IT works space consisting of computers and a whiteboard, and a hallway that leads to a chemical laboratory. The chemical lab was equipped with a typical apparatus like beakers, furnaces, test tubes, the periodic table, etc., to demonstrate simple experiments. Additionally, we also provided a model of ethanol molecular structure to showcase explanations involving 3D models.

Avatar representation was implemented with the help of a free asset model from unity Asset store (<https://assetstore.unity.com/>). Inverse kinematics was applied to the model to provide natural movement while interacting in the virtual environment. In the future, we would explore the ability to change the character model and the impact on social presence.

4 USE CASES

VR provides opportunities to deliver technical education in ways not possible through traditional methods, thereby broadening the range of tools available to modern educators, in order to reach more students and subjects[3]. Taking this into consideration, we opted a scenario where a professor explains the molecular structure and creation of ethanol to demonstrate the working of the prototype

system. The professor explains the chemical formula on the white-board while positioning the camera in an ideal location and later moves to the lab area. In the lab area, after a short tour, the creation process is explained with a 3D structure model of the end product. The features demonstrated can be applied to various industries and fields like medicine, surgery, building space tours, and remote training.

5 USER EXPERIENCE AND DISCUSSION

We created a system prototype that can be classified into Virtual Camera, Desktop Mirroring, and Virtual Environment. We selected a chemistry lab experience that would showcase the abilities of the system. From initial application testing and informal feedback, we found that users prefer a hybrid method to control the camera placement and prefer attaching the desktop view to the virtual camera model. We would explore this with a formal user study assessing the modes of interactivity for the virtual camera and the desktop-mirroring tablet. We are particularly interested in understanding how manipulating the camera and tablet affects the user joining from VR. We will conduct user tests for the following scenarios: Manual – here the presenter in VR controls the position and location of the camera and tablet by grabbing and moving them manually; Automatic – here the system automatically moves the camera feed to make sure the object of interest remains in focus, and the tablet displaying the other participants remains in view; Hybrid – the presenter can switch between automatic and manual modes dynamically. We are interested in understanding how these conditions will affect user experience as well as task load.

6 CONCLUSION

We presented an MR solution that can be integrated into existing teleconferencing solutions like Zoom, Skype, hangouts, etc., to provides a three-dimensional stage within 2D conferences. With XRTB, users can visit places and manipulate objects that might not be possible in the physical world while taking webcam participants along for the ride. We broke down the system into three parts(virtual camera, Desktop mirroring, and virtual environment) and described the motivation behind the development. We discuss features, modes of interaction, our findings from development, and plans for further development and user testing.

REFERENCES

- [1] Huidong Bai, Prasanth Sasikumar, Jing Yang, and Mark Billinghurst. 2020. A User Study on Mixed Reality Remote Collaboration with Eye Gaze and Hand Gesture Sharing. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [2] Martin Bauer, Gerd Kortuem, and Zary Segall. 1999. "Where are you pointing at?" A study of remote collaboration in a wearable videoconference system. In *Digest of Papers. Third International Symposium on Wearable Computers*. IEEE, 151–158.
- [3] J. T. Bell and H. C. Fogler. 2004. The application of virtual reality to (chemical engineering) education. In *IEEE Virtual Reality 2004*. 217–218. <https://doi.org/10.1109/VR.2004.1310077>
- [4] Mark Billinghurst and Hirokazu Kato. 2002. Collaborative augmented reality. *Commun. ACM* 45, 7 (2002), 64–70.
- [5] Henry Chen, Austin S Lee, Mark Swift, and John C Tang. 2015. 3D collaboration method over HoloLens™ and Skype™ end points. In *Proceedings of the 3rd International Workshop on Immersive Media Experiences*. 27–30.
- [6] Pavel Gurevich, Joel Lanir, Benjamin Cohen, and Ran Stone. 2012. TeleAdvisor: a versatile augmented reality tool for remote assistance. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 619–622.
- [7] Shunichi Kasahara, Valentin Heun, Austin S Lee, and Hiroshi Ishii. 2012. Second surface: multi-user spatial collaboration system based on augmented reality. In *SIGGRAPH Asia 2012 Emerging Technologies*. 1–4.
- [8] Hannes Kaufmann and Dieter Schmalstieg. 2002. Mathematics and geometry education with collaborative augmented reality. In *ACM SIGGRAPH 2002 conference abstracts and applications*. 37–41.
- [9] Thammathip Piumsomboon, Youngho Lee, Gun Lee, and Mark Billinghurst. 2017. CoVAR: a collaborative virtual and augmented reality system for remote collaboration. In *SIGGRAPH Asia 2017 Emerging Technologies*. 1–2.
- [10] Lauriane Pouliquen-Lardy, Isabelle Milleville-Pennel, Francois Guillaume, and Franck Mars. 2016. Remote collaboration in virtual reality: asymmetrical effects of task distribution on spatial processing and mental workload. *Virtual Reality* 20, 4 (2016), 213–220.
- [11] Harrison Jesse Smith and Michael Neff. 2018. Communication behavior in Embodied virtual reality. In *Conference on Human Factors in Computing Systems - Proceedings*, Vol. 2018-April. Association for Computing Machinery, New York, New York, USA, 1–12. <https://doi.org/10.1145/3173574.3173863>
- [12] Rajinder S Sodhi, Brett R Jones, David Forsyth, Brian P Bailey, and Giuliano Maciocci. 2013. BeThere: 3D mobile collaboration with spatial input. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 179–188.
- [13] Tzu Yang Wang, Yuji Sato, Mai Otsuki, Hideaki Kuzuoka, and Yusuke Suzuki. 2020. Effect of body representation level of an avatar on quality of AR-based remote instruction. *Multimodal Technologies and Interaction* 4, 1 (mar 2020). <https://doi.org/10.3390/mti4010003>